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Background and Motivation

Regarding the design problem for SSD Lasso recovery, we have already had considerable discussion. We centered the random vector $\mathbf{u}$ in the event S_λ and performed a Cholesky decomposition on its covariance matrix Σ , thereby shifting the complexity from the density to the integration region. However, this caused the integration upper limits to contain random variables and to be coupled with one another, and many elements within the upper limits are also coupled, making the problem difficult to solve. We attempted to propose some criteria, but the results were not satisfactory. We also conjectured that, with other v_j fixed, the probability (integral value) $P(S_\lambda)$ is monotonically increasing with respect to $\frac{v_i}{\gamma_{ii}}$, and tried to prove it. But we seem unable to “fix the other components” because the components are coupled. After that, we noticed that concentrating the complexity in the density seems easier to handle. We will temporarily depart from the Cholesky decomposition approach and try to provide a more rigorous line of thought from other perspectives.

Alternative Representation of the Probability

First, fix the active set, the signs, the effect sizes, $\mathbf{V}_{\mathcal{A}}^{\frac{1}{2}}$ and λ . Define

$$\Sigma = \mathbf{Z}_{\mathcal{A}} \mathbf{C}_{\mathcal{A}}^{-1} \mathbf{Z}_{\mathcal{A}}, \quad \mathbf{D} = \Sigma^{-1} = \mathbf{Z}_{\mathcal{A}} \mathbf{C}_{\mathcal{A}} \mathbf{Z}_{\mathcal{A}}^T = \sqrt{n} \lambda, \quad \mathbf{b} = \sqrt{n} \mathbf{V}_{\mathcal{A}}^{\frac{1}{2}} |\beta_{\mathcal{A}}|.$$

Here $\mathbf{D}$ is a positive definite matrix satisfying $\text{diag}(\mathbf{D}) = \mathbf{1}$.

Let $\mathbf{R} = -\mathbf{u}$. Then

$$\mathbf{R} \sim N_k(-t \Sigma \mathbf{1}, \Sigma), \quad P(S_\lambda) = P(\mathbf{R} > -\mathbf{b}).$$

The probability can be expressed as

$$P_S(\mathbf{D}) = \frac{|\mathbf{D}|^{\frac{1}{2}}}{(2\pi)^{\frac{k}{2}}} \int_{\mathbf{r} > -\mathbf{b}} \exp\left(-\frac{1}{2}(\mathbf{r} + t \mathbf{D}^{-1} \mathbf{1})^T \mathbf{D}(\mathbf{r} + t \mathbf{D}^{-1} \mathbf{1})\right) d\mathbf{r},$$

that is,

$$P_S(\mathbf{D}) = \frac{|\mathbf{D}|^{\frac{1}{2}}}{(2\pi)^{\frac{k}{2}}} \exp\left(-\frac{t^2}{2} \mathbf{1}^T \mathbf{D}^{-1} \mathbf{1}\right) \int_{\mathbf{r} > -\mathbf{b}} \exp\left(-\frac{1}{2} \mathbf{r}^T \mathbf{D} \mathbf{r} - t \mathbf{1}^T \mathbf{r}\right) d\mathbf{r}. \quad (1)$$

In this way, the integration region no longer changes with $\mathbf{D}$, and all the complexity is concentrated in the density.

Local Optimality Analysis

Denote $\ell(\mathbf{D}) = \log P_S(\mathbf{D})$, $\mathbf{a} = \mathbf{D}^{-1} \mathbf{1}$, and the conditional second-order raw moment $M(\mathbf{D}) = \mathbb{E}_{\mathbf{D}}[\mathbf{R} \mathbf{R}^T | \mathbf{R} > -\mathbf{b}]$.

Differentiating directly from Eq. (1), we obtain

$$\nabla_{\mathbf{D}} \ell(\mathbf{D}) = \frac{1}{2}(\mathbf{D}^{-1} + t^2 \mathbf{a} \mathbf{a}^T - M(\mathbf{D})). \quad (2)$$

Since the diagonal entries of the correlation matrix are fixed, the stationary point condition is

$$M(\mathbf{D})_{ij} = (\mathbf{D}^{-1})_{ij} + t^2 \mathbf{a}_i \mathbf{a}_j, \quad i \neq j. \quad (3)$$

Take any admissible symmetric perturbation

$$\mathbf{H} = \mathbf{H}^T, \quad \text{diag}(\mathbf{H}) = \mathbf{0}.$$

Differentiating Eq. (1) again gives the second-order directional derivative

$$D^2\ell(\mathbf{D})[\mathbf{H}, \mathbf{H}] = \frac{d^2}{d\epsilon^2}\ell(\mathbf{D} + \epsilon\mathbf{H})|_{\epsilon=0} = -\frac{1}{2}\text{tr}(\mathbf{D}^{-1}\mathbf{H}\mathbf{D}^{-1}\mathbf{H}) - t^2\mathbf{a}^T\mathbf{H}\mathbf{D}^{-1}\mathbf{H}\mathbf{a} + \frac{1}{4}\text{Var}_{\mathbf{D}}(\mathbf{R}^T\mathbf{H}\mathbf{R}|\mathbf{R} > -\mathbf{b}). \quad (4)$$

Points satisfying Eq. (3) are stationary points. If, in addition, Eq. (4) is required to be strictly negative for all nonzero admissible perturbations, then the stationary point can be confirmed as a local maximum. If some direction makes Eq. (4) positive, then a sufficiently small perturbation along that direction will still increase the true probability.

For example, at the orthogonal point, let

$$\mathbf{D} = \mathbf{I}, \quad \tau_i = b_i - t, \quad h_i = \frac{\phi(\tau_i)}{\Phi(\tau_i)}.$$

At this point the components are independent, and

$$\mathbb{E}[R_i|R_i > -b_i] = \mathbb{E}[R_i + t|R_i + t > -(b_i - t)] - t = h_i - t.$$

Take the free correlation coefficient $d_{ij} = D_{ij}$, which can simultaneously change D_{ij} and D_{ji} . According to Eq. (2),

$$\left.\frac{\partial\ell}{\partial d_{ij}}\right|_{\mathbf{D}=\mathbf{I}} = 2\frac{\partial\ell}{\partial D_{ij}} = t(h_i + h_j) - h_i h_j = h_i h_j \left(\frac{t}{h_i} + \frac{t}{h_j} - 1\right). \quad (5)$$

This shows that the orthogonal design is indeed very likely not optimal. When $\frac{t}{h_i} + \frac{t}{h_j} > 1$, we have $\left.\frac{\partial\ell}{\partial d_{ij}}\right|_{\mathbf{D}=\mathbf{I}} > 0$, which means that increasing positive correlation will increase the probability. Conversely, when $\frac{t}{h_i} + \frac{t}{h_j} < 1$, decreasing positive correlation will increase the probability.

We can directly obtain part of Lemma 3 from the original paper. Taking $\mathbf{z}_{\mathcal{A}} = \mathbf{1}$, we have $\mathbf{Z}_{\mathcal{A}} = \mathbf{I}$, $\mathbf{D} = \mathbf{C}_{\mathcal{A}}$. Under the completely symmetric assumption of the original paper, $\mathbf{D} = (1 - c)\mathbf{I} + c\mathbf{J}$, where $c = 0$ corresponds to $\mathbf{D} = \mathbf{I}$. Assuming $\mathbf{V} = \mathbf{I}$ and $|\beta_{\mathcal{A}}| = \beta\mathbf{1}$, we have $b_i = \sqrt{n}\beta$, $\tau_i = \tau = \sqrt{n}(\beta - \lambda)$, and $h_i = h = \frac{\phi(\tau)}{\Phi(\tau)}$. In the completely symmetric space, $\forall i \neq j, d_{ij} = c$. Therefore

$$\frac{d\ell}{dc} = \sum_{i < j} \frac{\partial\ell}{\partial d_{ij}} \frac{\partial d_{ij}}{\partial c} = \sum_{i < j} \frac{\partial\ell}{\partial d_{ij}}.$$

At $c = 0$ (i.e. $\mathbf{D} = \mathbf{I}$), since $h_i = h$ for all i ,

$$\left.\frac{d\ell}{dc}\right|_{c=0} = \sum_{i < j} \left.\frac{\partial\ell}{\partial d_{ij}}\right|_{\mathbf{D}=\mathbf{I}} = \sum_{i < j} (t(h_i + h_j) - h_i h_j) = \binom{k}{2} h(2t - h).$$

Setting $\left.\frac{d\ell}{dc}\right|_{c=0} \geq 0$ gives $2t \geq h$, i.e.,

$$2\sqrt{n}\lambda \geq \frac{\phi(\sqrt{n}(\beta - \lambda))}{\Phi(\sqrt{n}(\beta - \lambda))}.$$

Since β and λ in the original paper already absorb $\sqrt{n}$, let $t = \lambda$ and $b = \beta$, this is equivalent to condition (18) in the original paper:

$$2\lambda \geq \frac{g(\tau)}{G(\tau)}, \quad \tau = \beta - \lambda.$$